

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Quiz - 1
 Duration: 25 minutes

Semester: Spring 2026
 Full Marks: 20

CSE 340: Computer Architecture

Name: <i>Solution</i>	ID:	Section:
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1.

Instruction Set:

Add
 Sub
 Ld
 Add
 Sd
 Sub
 Sll
 Add
 Sub
 Sub
 Ld
 Sd
 Sub
 Sll

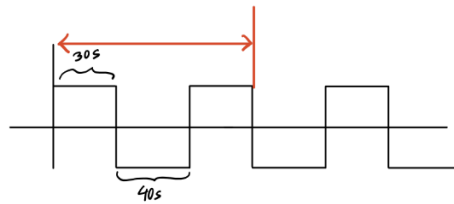


Figure : Add Instruction

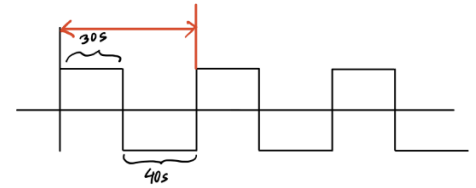


Figure : Sub Instruction

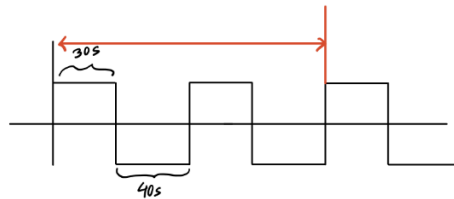


Figure : Ld Instruction

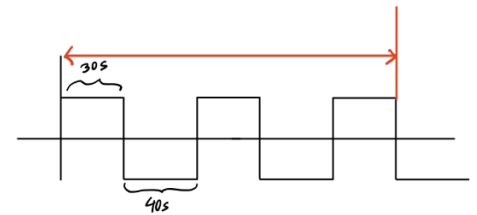


Figure : Sd Instruction

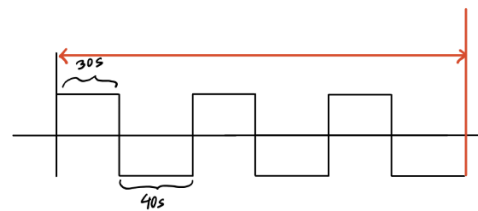


Figure : Sll Instruction

a. Find CPIs; You must answer in the following table. [1 x 5]

	Add	Sub	Ld	Sd	Sll
CPI	1.5	1	2	2.5	3
IC	3	5	2	2	2

b) Clock Cycle Time = $(30 + 40) \text{ ns}$
 $= 70 \text{ ns}$

Freq. = $\frac{1}{\text{Clock Cycle Time}}$

$= \frac{1}{70} \text{ Hz}$

$= 0.0142 \times 10^{-9} \text{ GHz}$

b. Find the Frequency in **GHz**. Answer: _____ [1]

c. Find the time to run the instruction set. Answer: _____ [4]

① Avg. CPI = $\frac{(3 \times 1.5) + (1 \times 5) + (2 \times 2) + (2 \times 2.5) + (3 \times 3)}{14} = \frac{24.5}{14} = 1.75$

CPU Time = $14 \times 1.75 \times 70 = 1715 \text{ ns}$

2. A tech company releases a new processor that can run existing software from older models without any changes. **Internally, the processor has been redesigned to improve efficiency, but none of the applications notice the difference.**

Write the name of the relevant idea from the 7 Great Idea(s) that is applied in the given scenario. [3]

Answer: **Use abstraction to simplify the design**

3. Suppose we run a program on a reference computer with a 4 GHz clock. Instruction count for that program is 15×10^9 and 2 clock cycles are needed to execute each instruction. CPI

a. Find the reference time. [3]

b. If the same program runs in 10s on your computer. Find the spec ratio and make a comment about the performance of your computer. [2+2]

Soln

$$\begin{aligned} \text{a) ref. time} &= \text{CPU time of ref. computer} \\ &= \frac{15 \times 10^9 \times 2}{4 \times 10^9} \\ &= 7.5 \text{ s} \end{aligned}$$

$$\left\{ \begin{array}{l} F_{\text{ref}} = 4 \text{ GHz} = 4 \times 10^9 \text{ Hz} \\ \text{CPI}_{\text{ref}} = 2 \\ \text{IC}_{\text{ref}} = 15 \times 10^9 \end{array} \right.$$

$$\text{b) CPU Time}_{\text{comp}} = 10 \text{ s}$$

$$\text{Spec Ratio} = \frac{7.5}{10} = 0.75$$

My computer runs the prog. 0.75 times slower than the ref. computer.